Information Matters: Sibling Influence on Study Choice

Stanislav Avdeev[1,3], Nadine Ketel[2],
Hessel Oosterbeek[1], Bas van der Klaauw[2]

[1]University of Amsterdam, [2]VU Amsterdam, [3]Tinbergen Institute

June, 2022

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)
  - What are the determinants of this decision?

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)
  - What are the determinants of this decision?
- Close peers, particularly siblings, seem to significantly influence study choice decisions (Almeida et al., 2021; Aguirre a and Matta, 2021; Goodman et al. 2015)

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)
  - What are the determinants of this decision?
- Close peers, particularly siblings, seem to significantly influence study choice decisions (Almeida et al., 2021; Aguirre a and Matta, 2021; Goodman et al. 2015)
  - Why do siblings follow each other?

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)
  - What are the determinants of this decision?
- Close peers, particularly siblings, seem to significantly influence study choice decisions (Almeida et al., 2021; Aguirre a and Matta, 2021; Goodman et al. 2015)
  - Why do siblings follow each other?
- Two main mechanisms explaining sibling spillovers in study choice

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)
  - What are the determinants of this decision?
- Close peers, particularly siblings, seem to significantly influence study choice decisions (Almeida et al., 2021; Aguirre a and Matta, 2021; Goodman et al. 2015)
  - Why do siblings follow each other?
- Two main mechanisms explaining sibling spillovers in study choice
  - Preference effect (common educational preferences)

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)
  - What are the determinants of this decision?
- Close peers, particularly siblings, seem to significantly influence study choice decisions (Almeida et al., 2021; Aguirre a and Matta, 2021; Goodman et al. 2015)
  - Why do siblings follow each other?
- Two main mechanisms explaining sibling spillovers in study choice
  - Preference effect (common educational preferences)
  - Information effect (sharing information about study options)

Motivation

- Field of study is one of the first pivotal decisions a person makes which leads to heterogeneity in earnings (Altonji et al., 2016)
  - What are the determinants of this decision?
- Close peers, particularly siblings, seem to significantly influence study choice decisions (Almeida et al., 2021; Aguirre a and Matta, 2021; Goodman et al. 2015)
  - Why do siblings follow each other?
- Two main mechanisms explaining sibling spillovers in study choice
  - Preference effect (common educational preferences)
  - Information effect (sharing information about study options)
- This paper allows to test directly the role of preference and information effects by exploiting admission lotteries in the Netherlands

Policy relevance

- If a policy maker wants to increase enrollment into specific majors, then preference and information mechanisms have different policy recommendations

Policy relevance

- If a policy maker wants to increase enrollment into specific majors, then preference and information mechanisms have different policy recommendations
  - If the preference effect is prevailing, a policy maker cannot influence students' choices

Policy relevance

- If a policy maker wants to increase enrollment into specific majors, then preference and information mechanisms have different policy recommendations
  - If the preference effect is prevailing, a policy maker cannot influence students' choices
  - If the information effect is prevailing, informational provision programs aimed at encouraging more students into specific majors would have multiplicative effects

Policy relevance

- If a policy maker wants to increase enrollment into specific majors, then preference and information mechanisms have different policy recommendations
  - If the preference effect is prevailing, a policy maker cannot influence students' choices
  - If the information effect is prevailing, informational provision programs aimed at encouraging more students into specific majors would have multiplicative effects
- Our results suggest that the information mechanism dominates in sibling spillovers in study choice
  - 80% of sibling spillovers are due to information sharing, and the remaining 20% are due to common preferences

Institutional background

- Students who complete the academic track in secondary school can enroll in any Dutch university
- Until 1999, medical schools were only accessible through the nationwide centralised admission lottery
- Average chance of winning is 45%, with some differences by high school GPA

Data and sample

- Data from Statistics Netherlands
  - For all: gender, age, migration background, enrollment, and field of study
  - For lottery participants: year of participation, category, and outcome

Data and sample

- Data from Statistics Netherlands
  - For all: gender, age, migration background, enrollment, and field of study
  - For lottery participants: year of participation, category, and outcome
- Sample
  - Older siblings (OS's): lottery participants who applied to medicine, dentistry, and veterinary in 1988-1999
  - Younger siblings (YS's): all (half)-siblings with the age difference 1-20 years
- If multiple OS's participated in a lottery, we keep the oldest one

Descriptive statistics

Empirical strategy

$$Y_i = \delta D_i + X_i\beta + \alpha_i + U_i$$

- Y_i is an indicator function for a sibling-pair i showing if
  1. a YS is enrolled in the program where an OS has *applied to*
  2. a YS is enrolled in the program where an OS is *enrolled in*

Empirical strategy

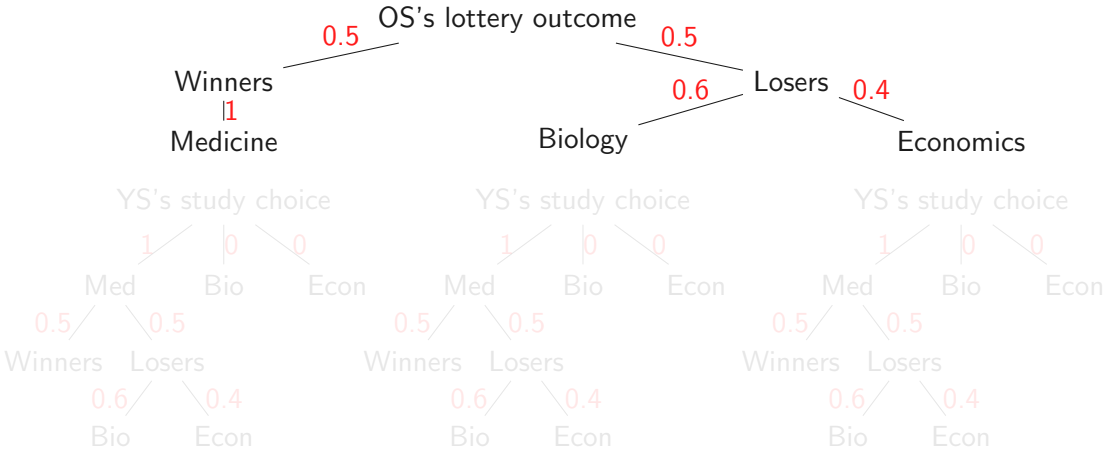
$$Y_i = \delta D_i + X_i\beta + \alpha_i + U_i$$

- Y_i is an indicator function for a sibling-pair i showing if
  1. a YS is enrolled in the program where an OS has *applied to*
  2. a YS is enrolled in the program where an OS is *enrolled in*
- D_i indicates the result of the first lottery of an OS
- X_i includes indicators for siblings' gender and non-western origin, age of an OS at the first lottery
- α_i are program, lottery category, year of the first lottery participation fixed effects, and their interactions
- Standard errors are clustered at the family level

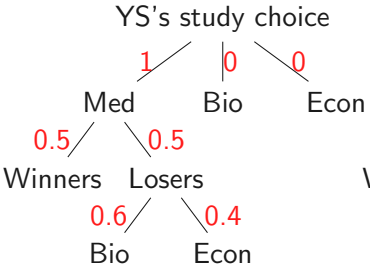
Hypotheses

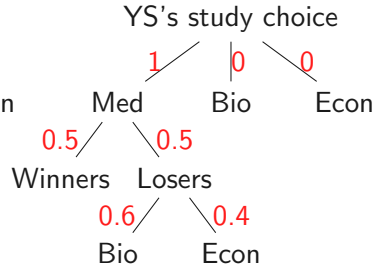
- Preference effect
  - Siblings shape educational preferences within families
- Information effect
  - OS's share information about the application process, study experience, group peers

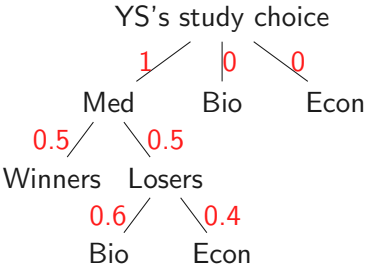
Predictions: Preference effect



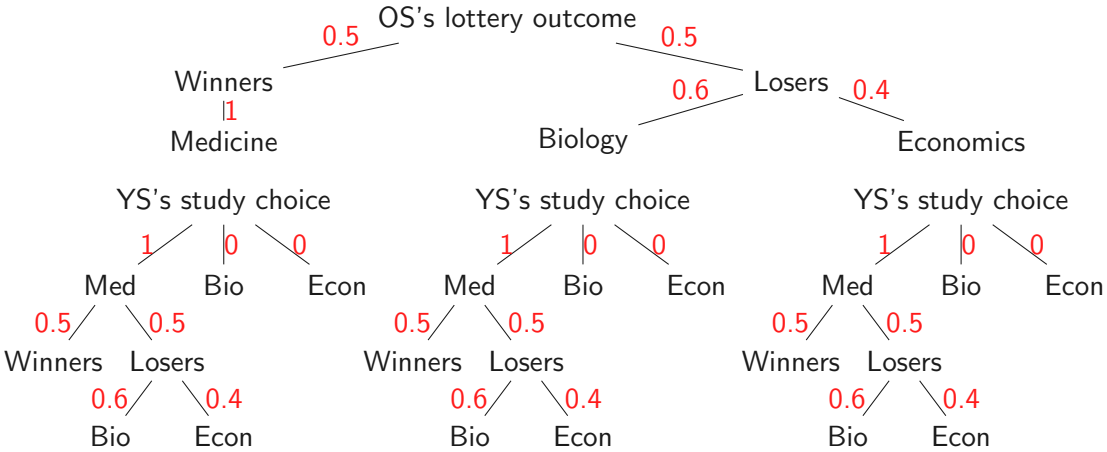
Predictions: Preference effect





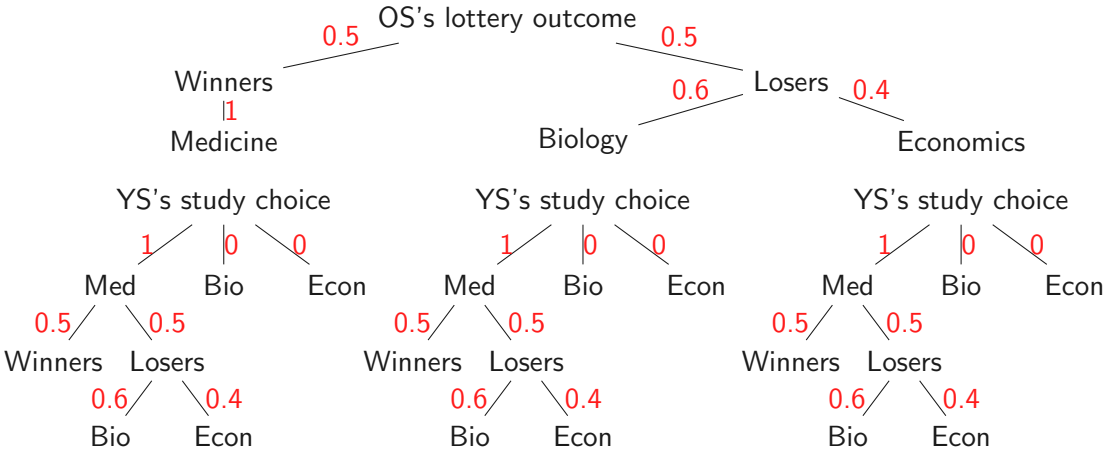


Predictions: Preference effect



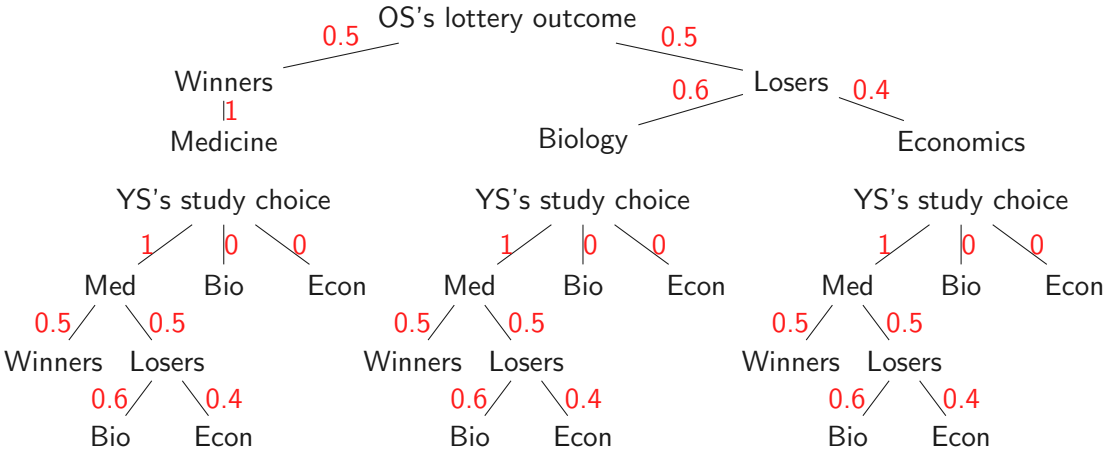
- $\Delta_{LP} = 0.5 - 0.5 = 0$

Predictions: Preference effect



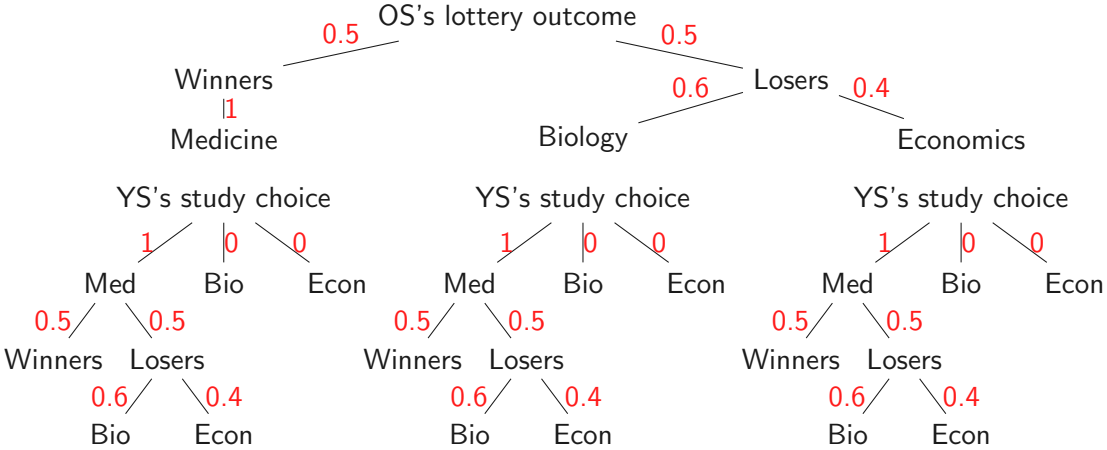
- $\Delta_{LP} = 0.5 - 0.5 = 0$

Predictions: Preference effect



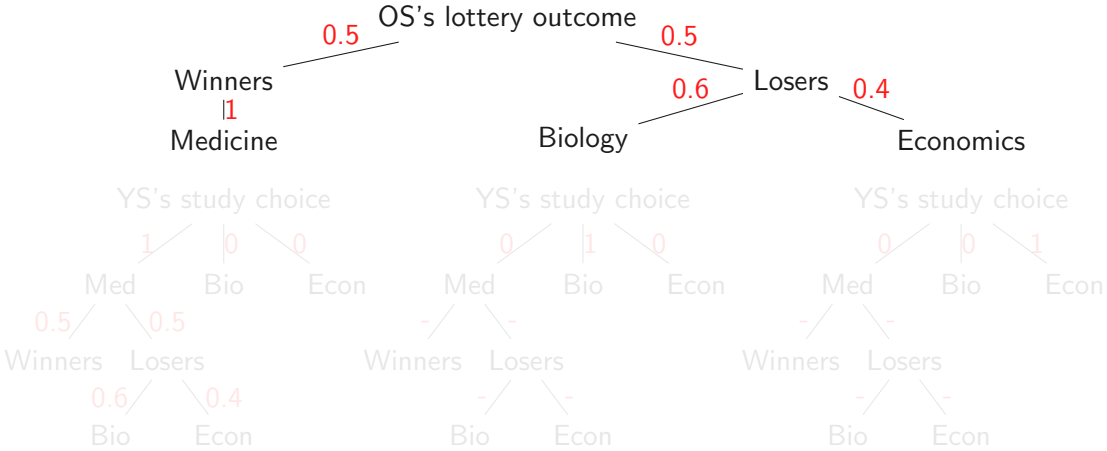
- $\Delta_{LP} = 0.5 - 0.5 = 0$
- $\Delta_{SP} = 0.5 - (0.6 * 0.5 * 0.6 + 0.4 * 0.5 * 0.4) = 0.24$

Predictions: Preference effect

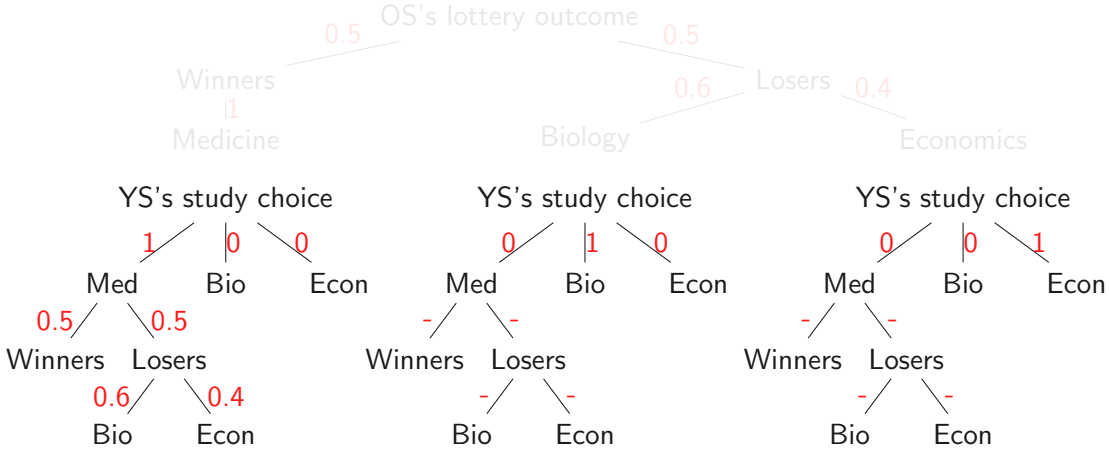


- $\Delta_{LP} = 0.5 - 0.5 = 0$
- $\Delta_{SP} = 0.5 - (0.6 * 0.5 * 0.6 + 0.4 * 0.5 * 0.4) = 0.24$

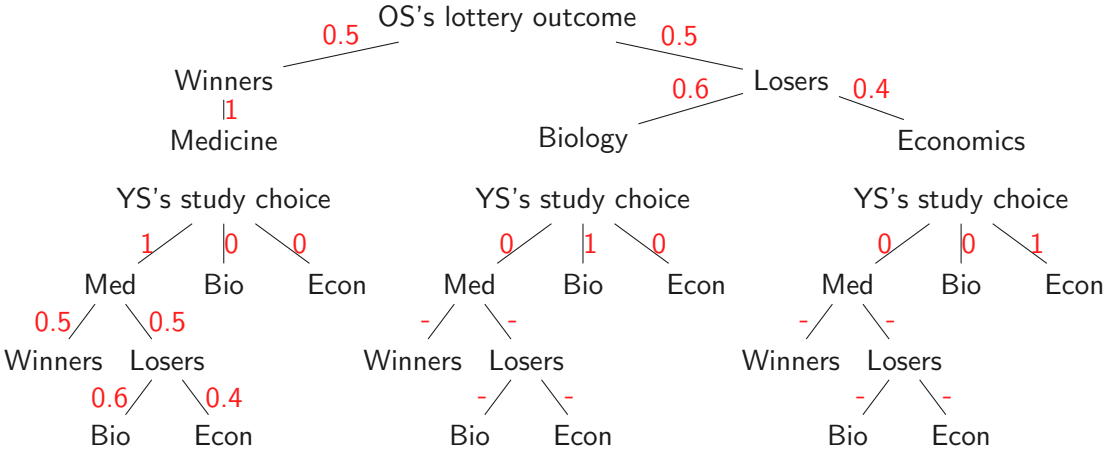
Predictions: Information effect



Predictions: Information effect

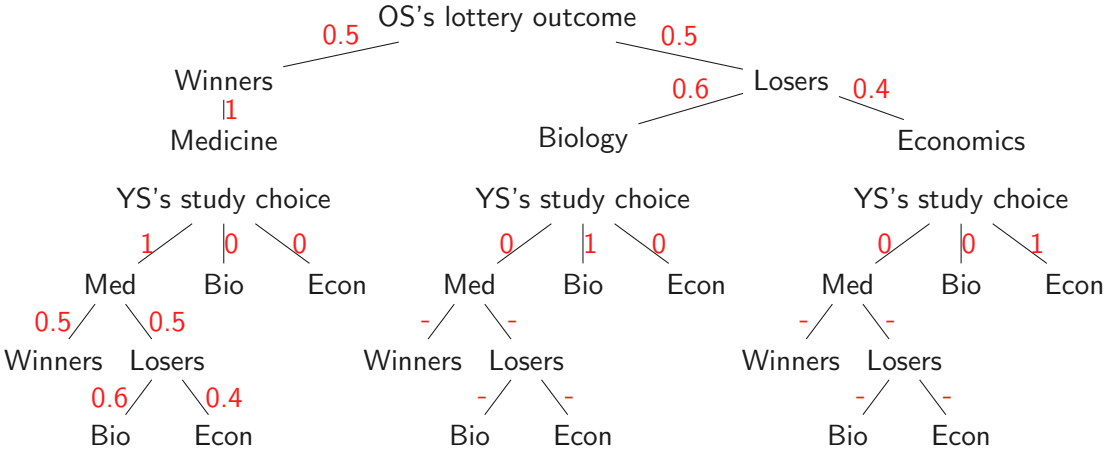


Predictions: Information effect



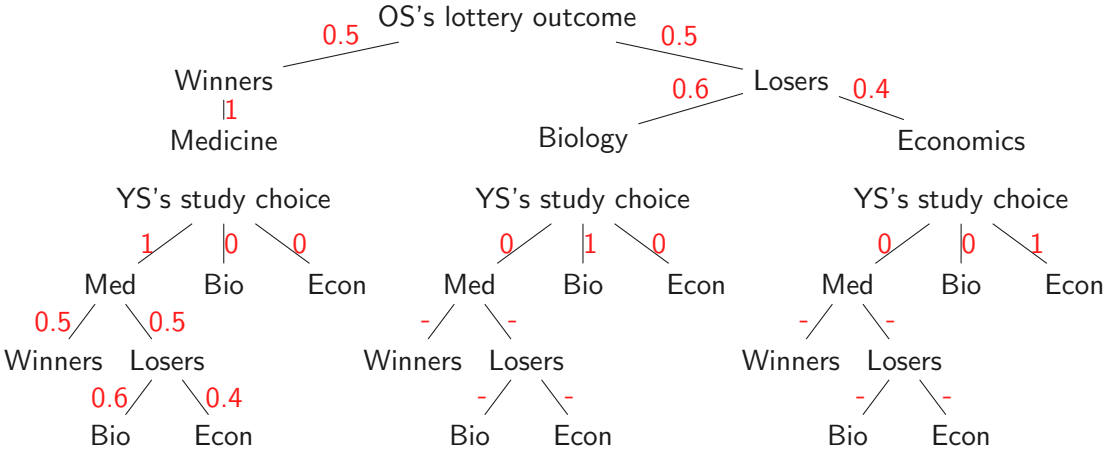
- $\Delta_{LP} = 0.5 - 0 = 0.5$

Predictions: Information effect



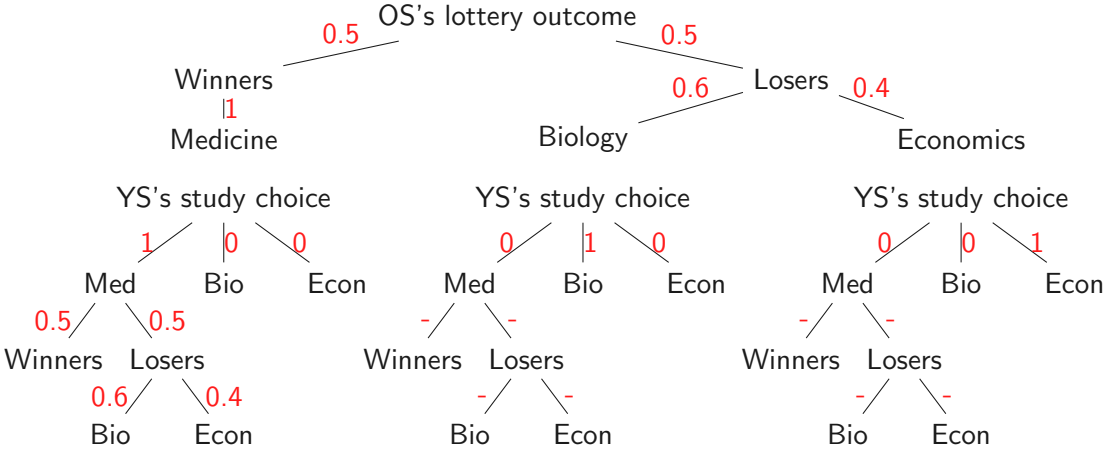
- $\Delta_{LP} = 0.5 - 0 = 0.5$

Predictions: Information effect



- $\Delta_{LP} = 0.5 - 0 = 0.5$
- $\Delta_{SP} = 0.5 - (0.6 + 0.4) = -0.5$

Predictions: Information effect



- $\Delta_{LP} = 0.5 - 0 = 0.5$
- $\Delta_{SP} = 0.5 - (0.6 + 0.4) = -0.5$

Predictions: Summary

Outcome	Preference	Information
Δ_{LP}	0	$+$
Δ_{SP}	$+$	$-$

- The stronger the preference effect, the *lower* Δ_{LP} and the *higher* Δ_{SP}
- The stronger the information effect, the *higher* Δ_{LP} and the *lower* Δ_{SP}

Results: Field of study choice

Outcome	YS is enrolled in	
	lottery program	same program
OS wins a lottery	0.032***	0.012***
s.e.	(0.003)	(0.004)
Control mean	0.032	0.056
Effect size	98.7%	21.6%
N	27642	27642

- As $\Delta_{LP} > \Delta_{SP}$, the information effect is prevailing

Results: University enrollment

Outcome	YS is enrolled in	
	university	same university
OS wins a lottery	0.004	0.005
s.e.	(0.006)	(0.006)
Control mean	0.810	0.207
Effect size	0.5%	2.4%
N	27642	27642

- OS's study choice does not affect university enrollment decision of YS's
  - No extensive margin effects
- YS's do not follow OS's to the same university
  - Information about a program and not a desire to study with your sibling matters

Heterogeneity

- Gender Results
- Age difference Results
- Parental income Results
- Abiity Results
- Migration background Results

Robustness checks

	Outcome	Oldest OS	Youngest OS	One OS	Two siblings	No age limit
1	Lottery program	0.032***	0.031***	0.029***	0.030***	0.031***
	s.e.	(0.003)	(0.003)	(0.003)	(0.006)	(0.003)
	Control mean	0.032	0.032	0.031	0.026	0.032
	Effect size	98.7%	96.4%	94.4%	115.0%	98.4%
2	Same program	0.012***	0.012***	0.010***	0.010	0.012***
	s.e.	(0.004)	(0.003)	(0.003)	(0.008)	(0.003)
	Control mean	0.056	0.056	0.055	0.059	0.055
	Effect size	21.6%	20.5%	18.3%	17.6%	22.2%
	N	27642	27642	26646	7195	28222

- The results are robust to different sample definitions

Simulation

- Let us simulate study choice of siblings using actual choices with a varying preference parameter

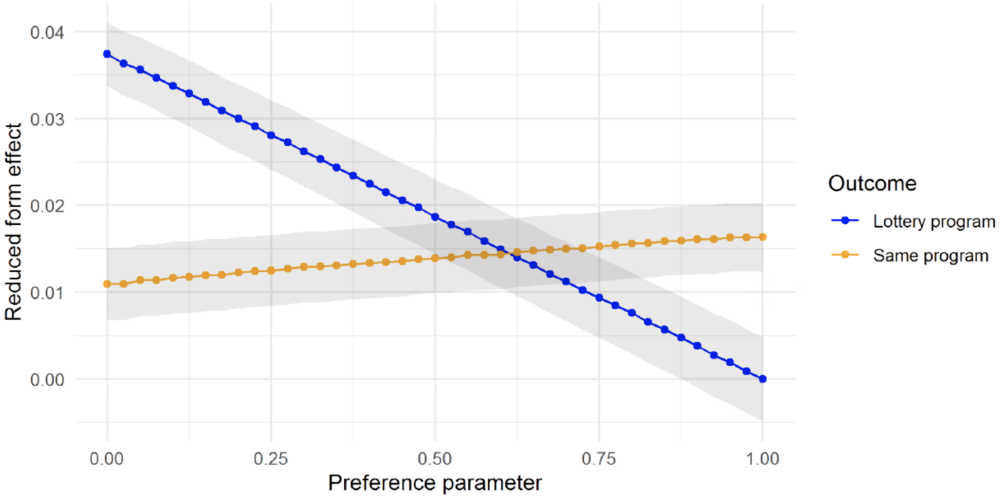
Simulation

- Let us simulate study choice of siblings using actual choices with a varying preference parameter
- Preference parameter – a probability of having similar study choices across YS's whose OS's are lottery winners and losers

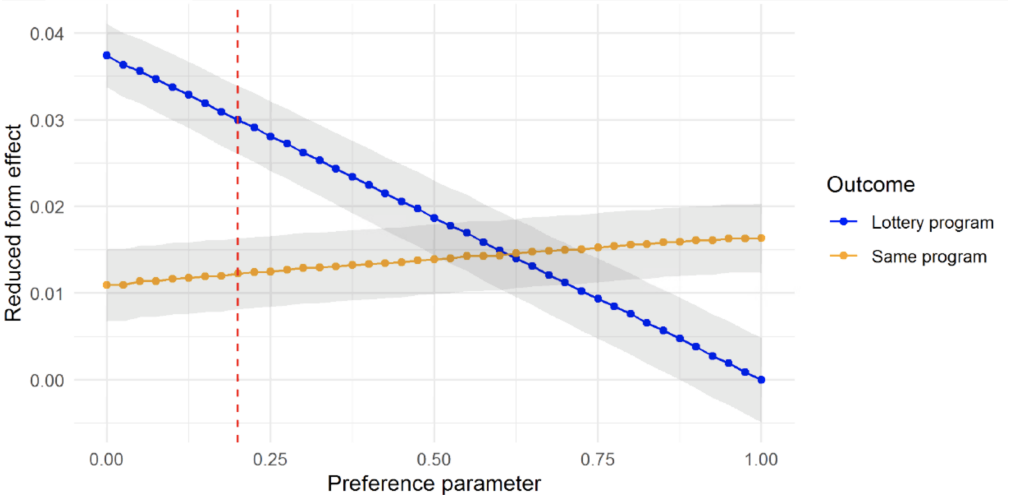
Simulation

- Let us simulate study choice of siblings using actual choices with a varying preference parameter
- Preference parameter – a probability of having similar study choices across YS's whose OS's are lottery winners and losers
  - when a preference parameter is 1, YS's study choice are identical across YS's whose OS's are lottery winners and losers
  - when a preference parameter is 0, YS's study choice are more similar to their OS's study choice

Simulation


0.04
0.03
0.02
0.01
0.00
Reduced form effect
0.00
0.25
0.50
0.75
1.00
Preference parameter
Outcome
Lottery program
Same program

Simulation



- A preference parameter is around 0.2 $\implies$ the information effect dominates

Conclusion

- Siblings influence each other's study choice
- They operate as information providers rather than preference changers
- Study choice spillovers operate solely through the information mechanism among
  - closely related siblings in terms of age
  - siblings from poorer families
- Peer networks and information sharing play a big role in determining study choice

Descriptive statistics: Personal characteristics

Older siblings

Outcome	Lottery winners	Lottery losers	p-value
Female	0.59	0.59	0.62
Non-western immigrant	0.07	0.07	0.39
Age at first application	18.82	18.86	0.74
Family size	2.97	2.97	0.42
N	8029	10002	

Younger siblings

Outcome	Winning siblings	Losing siblings	p-value
Female	0.49	0.48	0.02
Non-western immigrant	0.08	0.09	0.13
Age difference in years	4.50	4.43	0.35
Same sex pairs	0.49	0.50	0.21
N	12294	15348	

Go back

Descriptive statistics: Enrollment

Older siblings

Outcome	Lottery winners	Lottery losers
Enrolled in a university	0.99	0.95
Enrolled in a lottery program	0.95	0.44
N	8029	10002

Younger siblings

Outcome	Winning siblings	Losing siblings
Enrolled in a university	0.83	0.81
Enrolled in a lottery program	0.10	0.08
N	12294	15348

Go back

Heterogeneity: Gender

	Outcome	Male-Male	Female-Female	Male-Female	Female-Male
1	Lottery program	0.027***	0.042***	0.045***	0.022***
	s.e.	(0.008)	(0.007)	(0.009)	(0.005)
	Control mean	0.036	0.037	0.037	0.021
	Effect size	75.8%	113.9%	121.6%	105.9%
2	Same program	-0.005	0.015*	0.028**	0.011*
	s.e.	(0.009)	(0.008)	(0.009)	(0.006)
	Control mean	0.071	0.067	0.058	0.034
	Effect size	-6.6%	22.7%	48.7%	31.0%
	N	5798	7865	5621	8358

- The information effect dominates among same-sex siblings, especially brothers

Go back

Heterogeneity: Age difference

	Outcome	< 2	2-4	4-6	> 6
1	Lottery program	0.023***	0.031***	0.044***	0.035***
	s.e.	(0.007)	(0.005)	(0.010)	(0.007)
	Control mean	0.033	0.032	0.031	0.031
	Effect size	68.7%	97.8%	142.5%	112.5%
2	Same program	-0.001	0.012**	0.023**	0.018**
	s.e.	(0.008)	(0.006)	(0.010)	(0.007)
	Control mean	0.068	0.062	0.056	0.032
	Effect size	-1.8%	19.9%	41.3%	55.7%
	N	5857	10593	4702	6239

- The information effect dominates among siblings closer in age, as they can provide more relevant information

Go back

Heterogeneity: Parental income

	Outcome	< median	> median
1	Lottery program	0.021***	0.041***
	s.e.	(0.004)	(0.005)
	Control mean	0.019	0.043
	Effect size	108.3%	95.9%
2	Same program	0.002	0.020***
	s.e.	(0.005)	(0.006)
	Control mean	0.046	0.065
	Effect size	4.2%	31.0%
	N	13870	13871

- The information effect dominates among siblings from poorer families

Go back

Heterogeneity: Ability

	Outcome	A	B	C	D	E	F
1	Lottery program	-0.035	0.006	0.030**	0.032***	0.031***	0.030***
	s.e.	(0.102)	(0.021)	(0.014)	(0.007)	(0.007)	(0.005)
	Control mean	0.092	0.058	0.041	0.037	0.032	0.024
	Effect size	-37.7%	10.5%	72.8%	87.0%	98.0%	124.5%
2	Same program	-0.030	-0.008	0.021	0.003	0.015**	0.011*
	s.e.	(0.097)	(0.023)	(0.015)	(0.008)	(0.007)	(0.007)
	Control mean	0.112	0.071	0.050	0.058	0.056	0.055
	Effect size	-26.8%	-11.0%	42.8%	4.3%	26.4%	20.8%
	N	676	1942	2916	6435	7171	9154

- Higher ability OS's have no effect on YS's study choice (hard to keep up with overachievers?)

Go back

Heterogeneity: Migration background

	Outcome	Migrant	Non-migrant
1	Lottery program	0.064***	0.030***
	s.e.	(0.017)	(0.003)
	Control mean	0.022	0.033
	Effect size	291.7%	91.1%
2	Same program	0.044**	0.011***
	s.e.	(0.019)	(0.004)
	Control mean	0.051	0.056
	Effect size	85.4%	19.7%
	N	2360	25282

- Sibling spillovers are stronger among siblings with migration background

Go back